
Title:

Project 2: Pharmaceutical Economic Data Hub

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Project Description: This project involves building a robust, high-performance data infrastructure to track the complex economic landscape of pharmaceuticals. The team will be responsible for creating a "Source of Truth" database that aggregates disparate information regarding drug packaging, dosing frequency, and market pricing from various government and public sources. The challenge lies in data normalization—taking unstructured or messy data from different agencies and mapping it to a clean, unified schema. This system will essentially power the "Backend Logic" for future analytics, allowing the company to query specific attributes of a drug (e.g., "What is the unit cost for this specific package size?") with high reliability. It requires strong Data Engineering, SQL architecture, and API development skills.

Expected Deliverables:

- **Automated Ingestion Pipelines:** Python scripts to scrape/fetch data from public sources (NADAC, Orange Book, etc.) on a scheduled basis.
- **Normalized Database Schema:** A PostgreSQL (or similar) database designed to handle the many-to-many relationships between drug products, package sizes, and pricing tiers.
- **Data Quality Framework:** Automated tests to flag anomalies (e.g., if a drug's price drops by 90% overnight, flag it for review).
- **REST API:** A secure internal API endpoint that allows other applications to query drug economic data by NDC (National Drug Code).

Key Questions to Explore:

- How do we handle "schema drift" when a government source changes their file format without warning?
- What is the most efficient way to map drugs across different datasets that might use slightly different naming conventions?
- How can we design the database to provide historical tracking (Time-Travel queries) to see how data points changed over time?

Resources & Support:

- **Data:** Publicly available sources including **NADAC** (National Average Drug Acquisition Cost), **FDA Orange Book** (Approved Drug Products), **FDA Purple Book** (Biological Products), and **OpenFDA**.
- **Technical stack support:** As needed.

Potential Tasks:

1. The "Generic Substitution" Logic (Business Logic)

Instead of just listing prices, the system should actively suggest cheaper alternatives based on FDA Therapeutic Equivalence (TE) codes.

- **The Problem:** "This Brand Name drug is \$500. Is there a safe generic available?"
- **The Feature:**
 - Ingest **TE Codes** from the FDA Orange Book (e.g., "AB" rated means therapeutically equivalent).
 - Build a query engine that takes a high-cost NDC and returns the NDCs of all "AB-rated" generic equivalents sorted by lowest NADAC price.

2. CMS "Crosswalk" Mapping (Interoperability)

In the real world, drugs are bought by **NDC** (the box) but billed by **HCPCS/CPT** (the procedure). Mapping these is a nightmare for every healthcare company.

- **The Problem:** "I bought 100 units of this drug, but how do I bill Medicare for it?"
- **The Feature:**
 - Ingest the quarterly **CMS NDC-to-HCPCS Crosswalk** files (publicly available from CMS.gov).
 - Create a "Billing Translator" API: Input an NDC $\rightarrow$ Output the J-Code (HCPCS), the Billing Units, and the Medicare Allowable Payment Limit.

3. Price Volatility & Forecasting (Data Science)

Move from looking at *current* prices to predicting *future* risk.

- **The Problem:** "Drug X has been stable for years, but suddenly spiked 300%. We need to know which drugs are volatile."
- **The Feature:**
 - **Volatility Index:** Calculate a "Stability Score" for every drug. If a drug's price changes more than 5% month-over-month, flag it as "High Risk."
 - **Inflation Alerts:** Use historical NADAC data to identify drugs outpacing standard inflation.
 - **Forecasting (Stretch Goal):** Use a simple Time-Series model (like Facebook Prophet or ARIMA) to predict the price of a drug 6 months from now.